Skills Summary

Rigid Motions on the Coordinate Plane

Use this reference while completing your worksheet or when studying.

Skill 1 — Translations: slide every point

Rule: $(x, y) \to (x + h,\, y + k)$. Positive h moves right, negative h moves left; positive k moves up, negative k moves down.

Apply it to *every* vertex separately — the rule is about points, and a figure moves because all of its points do. Nothing rotates, nothing flips, and every length survives, because both endpoints of a segment shift by the same amount and the gap between them cannot change.

Example: Image of $A(-2, 5)$ under $(x, y) \to (x + 7,\, y - 4)$.
Step 1. The rule is already in coordinate form, so there is nothing to set up — work one coordinate at a time and let the signs do the moving.
Step 2. $x' = -2 + 7, \quad y' = 5 - 4.$
$\Rightarrow \quad A' = (\mathbf{5},\, \mathbf{1})$

Try it: Image of $B(3, -6)$ under the same rule.

check: $B' = (\mathbf{10},\, \mathbf{-10})$

Skill 2 — Reflections: which coordinate flips

Over the x-axis: $(x, y) \to (x,\, -y)$. **Over the y-axis:** $(x, y) \to (-x,\, y)$. **Over the line $y = x$:** $(x, y) \to (y,\, x)$.

The trick for keeping the first two apart: the axis you reflect *over* is the coordinate that stays put. A horizontal mirror flips you up and down, so it changes y; a vertical mirror flips you sideways, so it changes x.

The mirror is always the perpendicular bisector of the segment joining a point to its image — a free check on any answer.

Example: Reflect $C(4, -7)$ over the x-axis.
Step 1. The mirror is horizontal, so it can only change how far up or down the point is — that is the y-coordinate, and x is left alone.
Step 2. $x' = 4, \quad y' = -(-7).$
$\Rightarrow \quad C' = (\mathbf{4},\, \mathbf{7})$

Try it: Reflect $D(-5, 2)$ over the x-axis.

check: $D' = (\mathbf{-5},\, \mathbf{-2})$

Skill 3 — Rotations about the origin

$90°$ **counterclockwise:** $(x, y) \to (-y,\, x)$. $180°$**:** $(x, y) \to (-x,\, -y)$. $90°$ **clockwise:** $(x, y) \to (y,\, -x)$.

For the quarter turns: *swap the coordinates, then fix one sign.* Which sign is the only thing the direction changes, so decide the direction first and read the rule second.

A rotation about the origin keeps every point the same distance from the origin, so x^2+y^2 is unchanged — swapping and negating cannot alter a sum of squares.

Example: Rotate $E(3, 8)$ by $90°$ counterclockwise about the origin.
Step 1. A quarter turn always swaps the two coordinates; counterclockwise is the direction that makes the new first coordinate the negated one.
Step 2. Swap to get $(8, 3)$, then negate the first: $x' = -8, \quad y' = 3$.
$\Rightarrow \quad E' = (\mathbf{-8},\, \mathbf{3})$

Try it: Rotate $F(-2, 6)$ by $90°$ counterclockwise about the origin.

check: $F' = (\mathbf{-6},\, \mathbf{-2})$

(!) Watch out: all three of these are **rigid** motions — they preserve every distance and every angle, so a figure and its image are always congruent. That is why you never need to measure all three sides to justify a congruence: naming the motion is the argument.